

HOMEWORK 2
DISCRETE MATHEMATICS
MATH 208, FALL 2022

DUE: 09/07/2022

CHAPTER 3 - PREDICATES AND QUANTIFIERS

1. Let h be *Ben is healthy.*, w : *Ben is wealthy.*, and s : *Ben is wise.* Express the following in English:
 - a) $h \wedge w$
 - b) $w \vee s$
 - c) $h \rightarrow (w \wedge s)$
 - d) $(h \rightarrow w) \wedge s$
2. Let $S(x, y)$ be the predicate x has seen y where the domain of discourse for x is all students in this class and the domain of discourse for y is all movies. Express the following in logical symbols using quantifiers.
 - a) Every student in this class has seen *Gone With The Wind*.
 - b) No student in this class has seen *Jaws*.
 - c) *The Shape of Water* has been seen by someone in this class.
 - d) Some students have seen every movie.
 - e) For each movie, there is at least one student in the class who has seen that movie.
3. Negate the propositions in Problem 1 in English.
4. Negate the propositions in Problem 2 in symbols. Note: An easy way to do this is to simply write $\neg$ in front of the answers in Problem 2. Don't do that! Give the negation with no quantifiers coming after a negation symbol.
5. Negate the propositions in Problem 2 in English. Note: An easy way to do this is to simply write *It is not the case that* in front of each proposition. Don't do that! Give the negation as a reasonably natural English sentence.

CHAPTER 4 - RULES OF INFERENCE

1. Show that $p \rightarrow q$ and $\neg p$, $\therefore \neg q$ is not a valid rule of inference. It is called the **Fallacy of denying the hypothesis**.
2. Prove the following symbolic argument is valid.

$$\begin{array}{l} \neg p \wedge q \\ r \rightarrow p \\ \neg r \rightarrow s \\ s \rightarrow t \\ \hline \therefore t \end{array}$$

3. Prove the following symbolic argument is valid.

$$\begin{array}{l} p \vee q \\ q \rightarrow r \\ (p \wedge s) \rightarrow t \\ \neg r \\ \neg q \rightarrow (p \wedge s) \\ \hline \therefore t \end{array}$$

4. Prove the following symbolic argument is valid.

$$\begin{array}{l} (\neg p \vee q) \rightarrow r \\ s \vee \neg q \\ \neg t \\ p \rightarrow t \\ (\neg p \wedge r) \rightarrow \neg s \\ \hline \therefore \neg q \end{array}$$

5. Express the following argument in symbolic form and prove the argument is valid. *If Ralph doesn't do his homework or he doesn't feel sick, then he will go to the party and he will stay up late. If he goes to the party, he will eat too much. He didn't eat too much. So Ralph did his homework.*
6. In Problem 5, show that you can logically deduce that Ralph felt sick.
7. In Problem 5, can you logically deduce that Ralph stayed up late?
8. Prove the following symbolic argument:

$$\begin{array}{l} \exists x(A(x) \wedge \neg B(x)) \\ \forall x(A(x) \rightarrow C(x)) \\ \hline \therefore \exists x(C(x) \wedge \neg B(x)) \end{array}$$